Linear Programming Course Summary

Generated by Scribe.AI

December 4, 2024

1 Key Terms

1.1 **Linear Programming Fundamentals

- **Slack Variable**: A variable added to a less-than-or-equal-to inequality constraint to transform it into an equality constraint. It represents the difference between the left-hand side and right-hand side of the inequality.

1.2 **Solution Space Analysis

- **Feasible Region**: The set of all points that satisfy all the constraints of a linear programming problem.
- **Vertex**: A point where two or more constraint boundaries intersect in the feasible region of a linear programming problem. In the Simplex method, optimal solutions are found at vertices.

1.3 **Simplex Method Algorithms

- **Basic Variable**: In the Simplex method, a variable that is expressed in terms of non-basic variables in a dictionary. In a basic feasible solution, basic variables are non-negative.
- **Non-basic Variable**: In the Simplex method, a variable set to zero in a given iteration of the algorithm. The values of the basic variables are determined by the values of the non-basic variables.
- **Pivot Operation**: The process of exchanging a basic and a non-basic variable in the Simplex method, leading to a new basic feasible solution.
- **Dictionary**: A representation of a linear programming problem where basic variables are expressed as functions of non-basic variables. It is used in the Simplex method to systematically move from one vertex to another.
- **Auxiliary Problem**: A modified linear programming problem introduced to find an initial feasible solution when the origin is not feasible in the original problem. It involves adding a new variable to shift the feasible region.

2 Problem Types

2.1 **Linear Programming Fundamentals

- **Finding an Initial Feasible Solution**: The Simplex method requires an initial feasible solution. If the origin is not feasible, an auxiliary problem is introduced to find a feasible solution to the original problem.
- **Using Dictionaries to Represent Solutions**: The Simplex method uses dictionaries to represent the current solution and to perform pivot operations efficiently. Each dictionary corresponds to a vertex of the feasible region.

2.2 **Advanced Linear Programming

- **Finding an Initial Feasible Solution**: The Simplex method requires an initial feasible solution. If the origin is not feasible, an auxiliary problem is introduced to find a feasible solution to the original problem.
- **Handling Infeasible Origins**: The origin $(0, 0)$ may not satisfy all constraints in a linear programming problem. Techniques like introducing an auxiliary problem are used to find a feasible starting point.
- **Optimizing in Higher Dimensions**: The Simplex method is extended to linear programming problems with more than two variables, requiring iterative pivoting operations to find the optimal solution.
- **Using Dictionaries to Represent Solutions**: The Simplex method uses dictionaries to represent the current solution and to perform pivot operations efficiently. Each dictionary corresponds to a vertex of the feasible region.

3 Algorithm Solutions

3.1 **Linear Optimization Techniques

- **Simplex Method**: Iteratively move from one vertex of the feasible region to another by setting non-basic variables to zero and choosing a new set of (non)basic variables to improve the objective function value. This is equivalent to moving from vertex to vertex in the feasible region.
- **Auxiliary Problem**: When the origin is not feasible, introduce a new variable x_0 and modify the constraints to create an auxiliary problem that maximizes $-x_0$. Solve this auxiliary problem using the Simplex method. If the optimal solution has $x_0 = 0$, then the solution is feasible for the original problem.